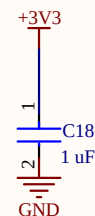
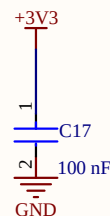
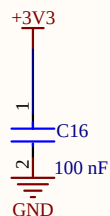
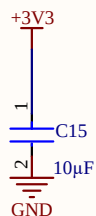


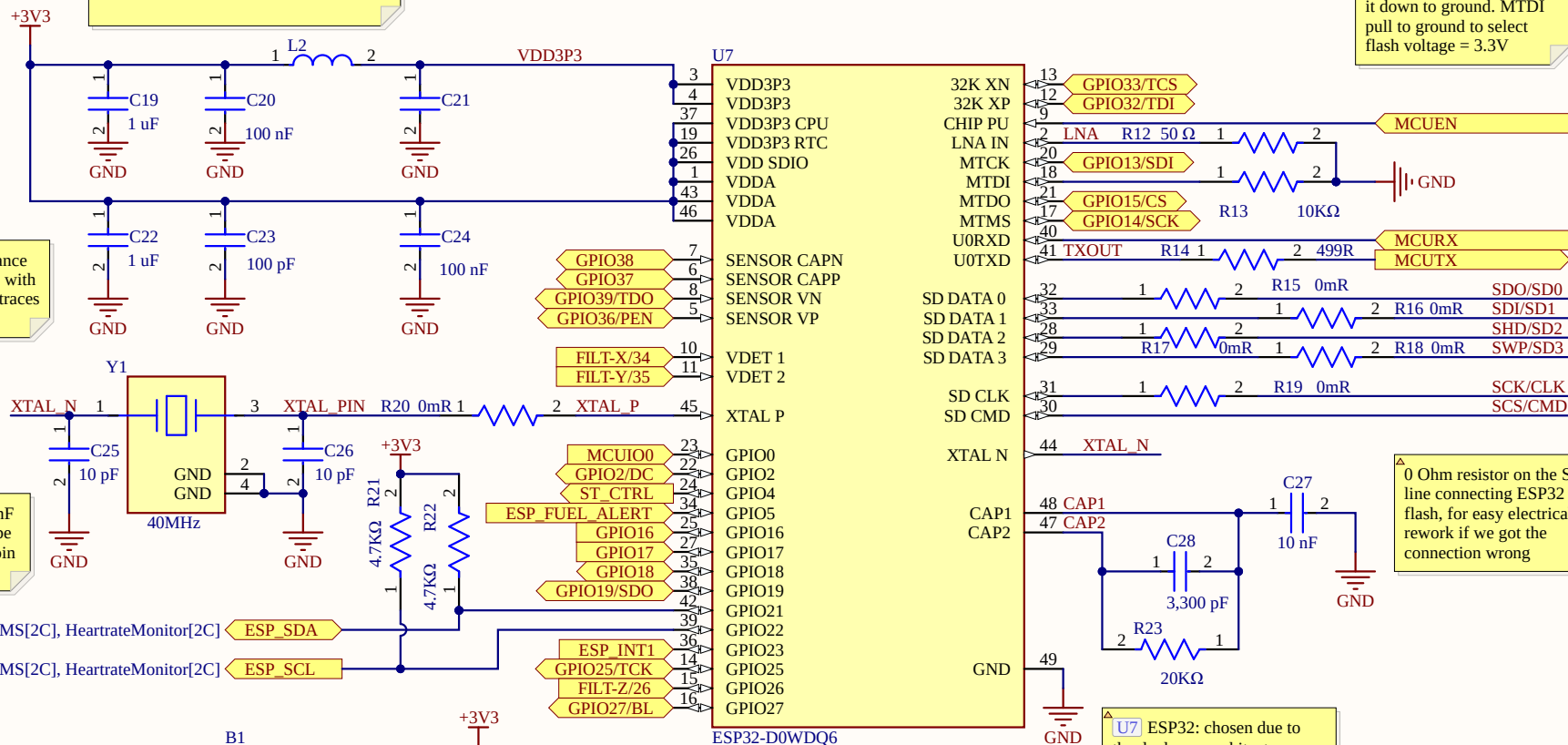
▲ Pi Filter on VDD3P3: suggested by espressif to block high frequency harmonics (around 11.3Mhz) from the MCU going into the 3V3 rail.

Equation:

$$f_c = 1/(2 \cdot \pi \cdot \sqrt{LC}) = 11.254\text{Mhz}$$



▲ antenna not used, so I pull it down to ground. MTDI pull to ground to select flash voltage = 3.3V



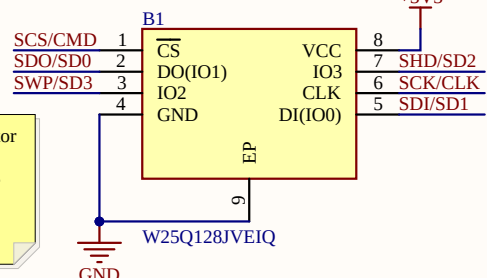
▲ Stray capacitance is around 3pF with short enough traces

▲ The second 100nF Cap above will be placed close to pin 1 VDDA

▲ 0 Ohm resistor on the SPI line connecting ESP32 to flash, for easy electrical rework if we got the connection wrong

Top Level[2C], BMS[2C], HeartrateMonitor[2C] ESP_SDA

Top Level[2C], BMS[2C], HeartrateMonitor[2C] ESP_SCL



▲ An external 40Mhz oscillator Y1 is used to provide a more accurate clock source for CPU (stable UART, accurate timing for pace identification using millis).

▲ B1 We need to use NOR flash for fast access of firmware stored in it, don't use NAND flash for this, which has higher storage capacity but cannot do random access well. We will use the W25Q128 with 16 megabyte of storage. This is a NOR flash working on 3.3V

▲ U7 ESP32: chosen due to the dual core architecture so we can use separate core for ADXL and graphics,

Title Microcontroller Subcircuit		
Size A	Number LUM-2026-SCH-006	Revision v1.0
Date: 5/22/2026	Sheet 6 of 9	
File: C:\Users\...\Microcontroller.SchDoc	Drawn By: Joe Nguyen	